INTERNATIONAL MONETARY FUND

CHAD

June 18, 2019

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION, FOURTH REVIEW UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT, REQUEST FOR MODIFICATION OF PERFORMANCE CRITERIA, AND FINANCING ASSURANCES REVIEW—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the Staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)

Chad Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgement	*No*

Chad's risks of external and overall debt distress are high but have nonetheless declined in the past year.[1] All but one external debt sustainability indicators are below their respective thresholds from 2019 onwards. The debt-to-revenue ratio moderately breaches its threshold under the baseline scenario. Overall, total public debt vulnerabilities are elevated although the present value (PV) of the public debt-to-GDP ratio remains on a downward trajectory. The debt sustainability analysis is based on projected continued fiscal prudence and an increase in non-oil revenues. Following the restructuring in 2018, the new Glencore debt contract has helped contain the impact of low oil prices on debt sustainability, as it allows for lower debt service when oil prices are lower.

[1] Chad's debt carrying capacity was rated weak according to the composite indicator (CI) based on the October 2018 WEO and the 2017 CPIA index.

PUBLIC DEBT COVERAGE

1. State and local debt has been added to the coverage of public debt. As in the previous DSA, coverage includes the central government, as well as state guaranteed external debt owed by the public oil company "Société des Hydrocarbures du Tchad" (SHT) (Text Table 1). This scope encompasses all public external debt; other public sector entities (including regions and other state-owned enterprises) do not have access to external financing. The Ministry of Finance plans to complete a census of public sector enterprises by the end of the year in order to assess the full amount of their outstanding domestic borrowing. Staff will use the findings to include SOE debt in the DSA.

2. The contingent liability stress test accounts for vulnerabilities associated with non-guaranteed state-owned enterprises (SOEs), unaudited domestic arrears, and financial markets (Text Table 1). Contingent liabilities from financial markets are set at 5 percent of GDP, which represents the average cost to the government of a financial crisis in a low-income country since 1980. The contingent liability stress test is customized to include domestic arrears that could potentially be validated by the ongoing audit. While there is significant uncertainty regarding these arrears, the size of the stress test is set at 8 percent, in addition to the standard amounts for SoEs' debt (2 percent of GDP) and financial market (5 percent of GDP).

Text Table 1. Chad: Coverage of Public-Sector Debt and Design of Contingent Liability Stress Tests

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central, state, and local governments, central bank		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	8	Domestic arrears could potentially be validated by an ongoing audit.
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2	
4	PPP	35 percent of PPP stock	0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)	15		

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND

A. Evolution and Composition of Debt

3. Chad's external public and publicly guaranteed (PPG) debt burden increased considerably over the past decade mainly on account of external commercial borrowings related to oil. Commercial borrowings (oil sale advances) from Glencore in 2013 to cover revenue shortfalls and in 2014 to purchase a share in the Doba Oil Consortium were the main contributors. Falling oil prices over 2014- 16 were the primary reason for reduced revenues available to repay oil sales advances. This debt has since been restructured twice, most recently in early 2018, which has considerably reduced its burden. At end- 2018, outstanding PPG external debt stood at about $2.6 billion (27 percent of GDP). Chad's CFAF-denominated debt held by the regional central bank (BEAC), the regional development bank (BDEAC), and bilateral creditors in the currency union (Republic of Congo, Equatorial Guinea, and Cameroon) amounts to 9.9 percent of GDP. It is not included in external debt, which is calculated on a currency basis.

4. The composition of external public debt has changed significantly over the past decade. The share of external debt from multilaterals has fallen sharply from about 87 percent in 2008 to 28 percent in 2018, while the share of commercial debt, which was virtually non-existent in 2008, is now trending down from a peak in 2017 of 54 percent, mostly to Glencore. Bilateral debt doubled over the decade but, as a share of total debt, it is still significantly less than commercial debt (Text Table 2). Consistent with the ECF arrangement, external debt is defined on a currency basis.

5. Domestic public debt has begun to decline in recent years (Text Table 3).[2] Following a peak in 2015, debt to the BEAC was restructured and Chad stopped borrowing from the BEAC. In addition to the debt owed to BEAC (33.7 percent of total debt), some debt denominated in local currency is held within the CEMAC region, including about 8.9 percent of total debt owed to official bilateral partners and BDEAC, and in the form of securities that could be held by non-resident banks. Since 2017, domestic debt has been declining as the authorities aim to loosen the bank-sovereign nexus and reduce domestic arrears.

[2] State and local debt amounts to less than 0.1 percent of GDP.

Table 1. Chad: External Debt Stock 2015–2019[1]

		2015	2016	2017	2018e
Total	(Millions of $)	2,735	2,622	2,649	2,671
	(Billions of CFA francs)	1,649	1,632	1,468	1,540
	(Percent of GDP)	25	27	25	25
	Billions of CFA francs				
Multilateral		375	390	385	422
	IMF	38	77	96	151
	World Bank/IDA	113	110	101	100
	African Development Fund/Bank	69	56	56	55
	Others	155	147	133	117
Bilateral		366	370	408	436
	Paris Club official debt	2	…	25	32
	Non-Paris Club official debt	364	370	383	404
	of which: China, People's Republic	144	156	132	130
	Libya	158	164	150	156
	India	27	30	27	21
Commercial[2]		875	862	705	664
	Share of Total (percent)				
Multilateral		23	24	26	28
Bilateral		23	23	26	27
Commercial[2]		54	53	48	45
	Memo: Millions of $				
Multilateral		635	631	695	738
	IMF	65	125	173	272
	World Bank/IDA	191	178	182	176
	African Development Fund/Bank	116	90	100	95
	Others	263	238	239	195
Bilateral		620	597	682	734
	Paris Club official debt	4	…	45	46
	Non-Paris Club official debt	616	597	637	689
	of which: China, People's Republic	244	252	226	226
	Libya	268	265	230	272
	India	46	49	38	37
Commercial[2]		1,481	1,394	1,272	1,199

Sources: Chadian authorities, selected creditors, and World Bank and IMF staff estimates.

[1]Includes only debt denominated in foreign currency.
[2]Glencore loan accounts for about 98 percent of commercial debt stock in 2017.

Text Table 2. Chad: Domestic Debt Stock 2015–2018

	2015	2016	2017	2018
Total (Billions of CFA francs)	**1191.8**	**1482.2**	**1445.6**	**1424.1**
(Percent of GDP)	**18.3**	**24.4**	**24.6**	**23.1**
Share of Total (in percent)				
Central Bank financing	38.1	33.3	33.2	33.7
Statutory advances [1]	*23.5*	*18.9*	...	...
Exceptional advance [1]	*11.7*	*11.5*	...	...
Consolidated debt	*2.9*	*3.0*	...	...
Commercial banks' loans	6.3	3.3	3.6	7.1
2011 Bond[2]	2.3	0.0	0.0	0.0
2013 Bond[2]	4.5	3.7	1.2	0.0
Treasury Bonds[3]	11.7	21.2	21.8	12.3
Treasury Bills	*7.0*	*11.2*	*11.7*	*20.6*
BDEAC	1.7	3.2	3.4	3.5
Republic of Congo	2.9	2.4	2.4	2.5
Equatorial Guinea	1.3	1.0	1.0	1.1
Cameroon	0.0	2.0	2.1	1.9
Domestic arrears	16.8	12.8	13.5	11.2
Others[4]	7.4	5.9	6.1	6.2

Source: Chadian authorities

[1]Includes advances that were consolidated in 2017.

[2]Issued through banks' syndication

[3]Auctionned in regional securities' market.

[4] Legal commitments, standing payment orders, and accounting arrears.

6. External payment arrears accumulated in 2016 and in 2017 but were reduced considerably in 2018. Due to liquidity challenges in 2016 and the first half of 2017, the government accrued external arrears vis-à-vis a number of multilateral, bilateral, and one commercial creditors (Mega bank from Taiwan province of China). At end-2017 about $102 million (about 1 percent of GDP) remained outstanding, mainly to bilateral creditors. The authorities have since reduced this stock to $63 million—particularly to the Rep. of Congo (about $55 million)—by paying the amount owed to the Islamic Development Bank and through a rescheduling agreement with Libya and India. Active discussions are underway to address all outstanding arrears, including with Angola, Equatorial Guinea, the Republic of Congo, the European Investment Bank, and Mega Bank. The authorities have taken concrete steps to prevent the further accumulation of arrears—including measures to improve coordination among relevant agencies and reactivate an escrow account for the payment of external debt at the BEAC.

B. Macroeconomic Forecast

7. The DSA's baseline scenario reflects policies and financing assumptions underlying the ECF arrangement and medium-term projections that reflect the Glencore debt restructuring. The growth projection has declined compared to the previous DSA (December 2018) from 4.6 and 6.1 percent in 2019 and 2020 respectively to 2.4 and 5.5 percent. The non-oil economic recovery is expected to be slower than originally projected due to the persistent impact of legacies from the crisis. The outlook assumes that the ongoing revenue-led fiscal consolidation will continue over the program horizon at a gradual pace and that spending control would be maintained. Export growth in 2019 is similarly expected to be weaker than previously expected despite sustained oil production because of a lower oil price forecast. Oil production is expected to continue to increase in the medium term, leading to higher oil revenues, higher exports and overall GDP growth (Figure 3).[3] The baseline scenario assumes full clearance of external arrears in 2019 and gradual repayment of domestic arrears.

Text Table 3. Chad: Macroeconomic Assumptions Comparison Table

	DSA December 2018		Current DSA	
	2017-22	2023-38	2018-23	2024-39
Real GDP (%)	3.5	3.2	4.1	3.1
Inflation (GDP deflator, %)	3.6	2.9	3.3	3.0
Average nominal interest rate on external debt (%)	3.5	1.4	2.9	1.5
Average nominal interest rate on domestic debt (%)	3.5	5.4	3.0	5.1
Primary Balance (% of GDP)	2.4	3.4	3.3	2.9
Real primary spending growth (%)	3.6	4.8	4.8	4.6
Exports of G&S growth (%)	13.3	2.6	12.3	2.3
Noninterest current account deficit (% GDP)	4.3	2.0	4.8	3.1

Sources: Chadian authorities and IMF staff estimates and projections.

8. Financing assumptions have been updated based on most recent information. Externally financed investment has remained unchanged, and the discount rate is kept at 5 percent over the forecast horizon. The grant element of new borrowing is assumed to decline gradually over the forecast horizon. With regards to domestic financing, based on a recent shift towards short term debt, the share of T-bills over the forecast horizon has been revised up. Reflecting this shift, average interest rate on domestic debt has been revised upward slightly.

9. The forecast is broadly realistic. The projected 3-year fiscal adjustment is in line with historical data on LIC adjustment programs. Continued fiscal prudence and efforts to raise non-oil revenues are expected to ensure a sustainable adjustment. The fiscal multiplier tool suggests that growth in 2020

[3] Note that the historical changes in Figure 3 are driven by the crisis and debt restructuring.

could be optimistic given the projected consolidation. However, staff expects a catalytic effect on growth from the program outside the immediate fiscal impulse. Private sector confidence is expected to be boosted by the improvement in fiscal health, improvement in budget execution, progress is identifying and clearing domestic arrears, and the authorities' ongoing efforts to implement the national development plan. This is consistent with expected private sector driven growth, led by private investment in the oil sector, as shown in Figure 4. New oilfield development projects have boosted expected private investment, while public sector investment remains low. The recent privatization of the cotton public enterprise is likely to help strengthen the private sector's contribution to growth.

C. Country Classification and Determination of Stress Test Scenarios

10. The composite indicator (CI) based on April 2019 World Economic Outlook (WEO) projections and an update of the CPIA index to 2019 levels indicates weak debt carrying capacity for Chad. The CI combines the CPIA score, external conditions as captured by world economic growth and country-specific factors. The April 2019 data indicate weak debt carrying capacity, reflecting mainly a low CPIA, very low remittances, and a low level of foreign reserves (Text Table 5).

Text Table 4. Chad: CI Score

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.677	1.03	42%
Real growth rate (in percent)	2.719	2.745	0.07	3%
Import coverage of reserves (in percent)	4.052	32.207	1.31	53%
Import coverage of reserves^2 (in percent)	-3.990	10.373	-0.41	-17%
Remittances (in percent)	2.022	0.000	0.00	0%
World economic growth (in percent)	13.520	3.559	0.48	19%
CI Score			2.48	100%
CI rating			Weak	

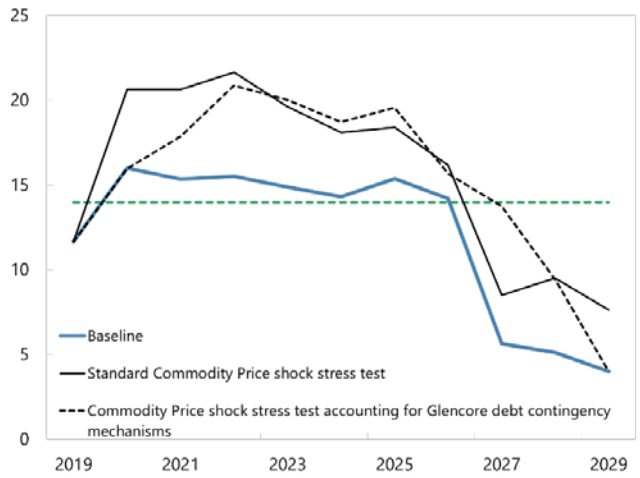
Source: IMF staff calculations. The CI cutoff for medium debt carrying capacity is 2.69.

11. The debt sustainability analysis relies on six standard stress tests and a customized oil price shock stress test (Figures 1 and 2 and Tables 3 and 4). The customized oil price shock entails oil prices lower than the baseline by 38 percent between 2020 and 2025 and has been calibrated to account for contingency mechanisms under the Glencore debt contract which limit the negative effect of the shock in the near term (Text Figure 1).[4] Debt service under the contract includes a mandatory amortization and interest payment plus a cash sweep component that falls to zero as the Doba oil price hits a certain price. The contract also allows Chad to shift out some of the mandatory payments as prices fall. In 2020 debt service will rise as the grace period on mandatory amortization expires under the contract. The contingencies significantly alleviate the debt service burden over the next few years,

[4] The fourth panel of Figure 1 presenting debt service-to-revenue ratios under standard alternative scenarios does not include the Glencore debt contract contingency mechanisms.

but moderately raises it starting in 2023. By 2029, the debt service level returns to the baseline as the Glencore debt is retired.

Text Figure 1. Chad: Debt Service-to-Revenue Ratios



Sources: Chadian authorities; IMF staff calculations.

DEBT SUSTAINABILITY

A. External Debt Sustainability

12. Public external debt is projected to gradually decline over the forecast horizon under the baseline scenario. Under the baseline scenario the present value of PPG external debt to GDP ratio, the present value of PPG external debt to exports ratio and the debt service to exports ratio are well below their thresholds (Figure 1). The debt service to revenue ratio is expected to drop below its threshold of 14 percent in 2019 and rise moderately above it over the medium term before dropping significantly as the Glencore debt matures. The projected path of this indicator has improved relative to its path in the last DSA.

13. Under stress tests, the thresholds are breached for all indicators. By increasing the future debt servicing burden, the Glencore debt contract raises the present value of PPG external debt, an effect that is most pronounced in the historical scenarios. Under the shock scenarios, the exports shock stress tests produce the most extreme scenario for all indicators except the debt service to revenue ratio, for which the commodity shock stress test is the most extreme. Under the export stress test, the threshold of the present value of PPG external debt-to GDP ratio is breached from 2021 until 2024. The present value of PPG external debt to exports ratio is breached from 2021 until 2026 and debt service to exports is breached from 2021 onwards.

B. Public Debt Sustainability

14. The benchmark for public debt is breached under the baseline. The PV of total public debt-to-GDP ratio projected at end-2019 stands at 41 percent, which is about 6 percentage points above the 35 percent benchmark level associated with heightened public debt vulnerabilities with a weak debt carrying capacity. The indicator achieves a below-threshold level by 2022 under the baseline scenario.

C. Risk Rating and Vulnerabilities

15. Chad's debt sustainability is less vulnerable to oil price fluctuations than before its debt restructuring. This reflects contingency mechanisms under the new Glencore debt contract, which allow lower external debt service to Glencore when oil prices are lower. As demonstrated by the recent oil price decline, the impact of a commodity price shock on debt sustainability is now limited. In 2019, external debt service to Glencore is now expected to be around 64 percent of what was scheduled before oil prices dipped late last year. Likewise, in a lower oil price scenario that accounts for the contingency mechanisms, the debt service to revenue ratio remains close to the baseline (Text Figure 1).

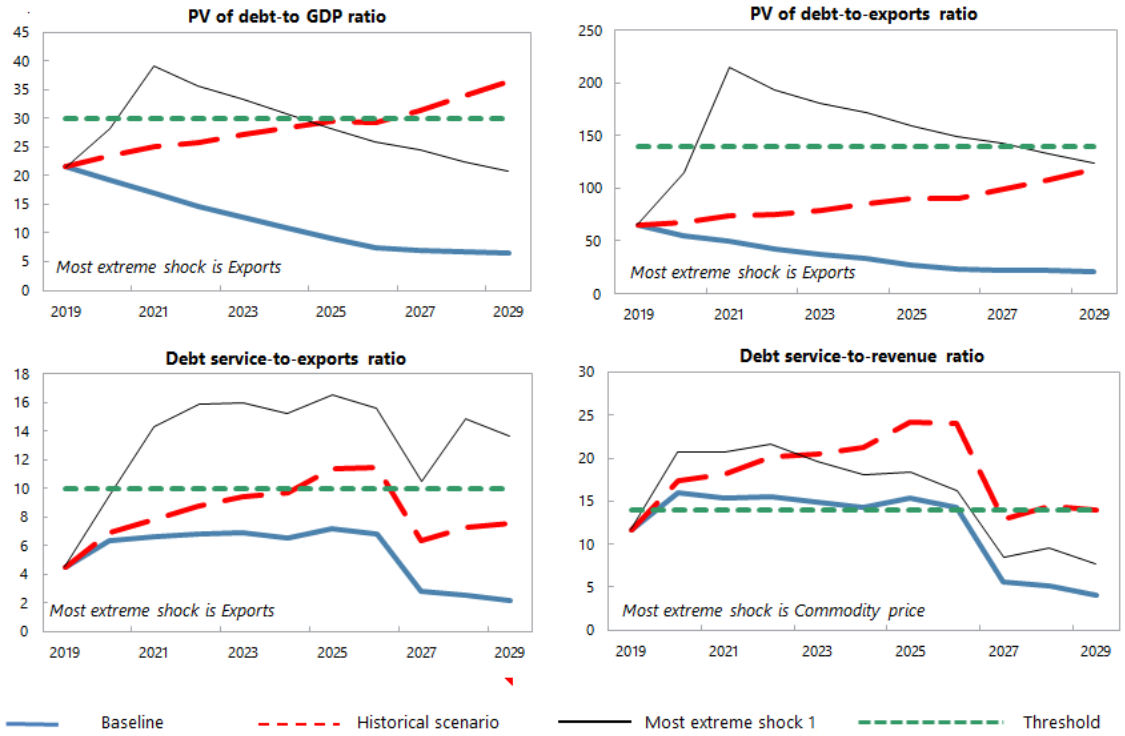
16. Chad is at high risk of external debt distress and high overall risk of public debt distress. Debt vulnerabilities have, however, declined significantly since the beginning of the program. The rescheduling of the Glencore debt along with the projected recovery in the oil sector and prudent fiscal policy result in debt burden indicators declining significantly over the near and medium terms. The projected path of the debt service to revenue indicator has improved significantly since the restructuring and in fact is better at the baseline oil price than at the higher price of the third review. Nonetheless, it remains moderately higher than the threshold under the new DSA framework. As such, Chad's external debt is assessed to be at high risk of debt distress. Additionally, the overall risk of debt distress is high based on the breach of an external debt sustainability indicator threshold and total public debt residing above its benchmark level. Mechanically, the CFAF-denominated debt held by the BEAC, BDEAC, and bilateral creditors would weaken the external debt sustainability indicators if the external DSA were done on residency basis. These claims do not face currency risk, and institutional ties with the creditors are relatively strong. Nonetheless, some difficulties may still be faced in restructuring such debt if necessary, and the risks associated with the rollover of securities held by non-resident (the scale of which is unclear) remains even if it is limited.

17. Significant efforts are warranted to ensure debt remains on a downward trajectory. Elevated vulnerabilities reinforce the need to maintain prudent fiscal policy including on external and domestic borrowing. While progress has been made recently to reduce the stock of external and domestic arrears, much more attention is needed going forward to clear the remaining arrears. Finally, continued effective inter-agency coordination to strengthen the capacity to record and monitor public debt is very important to better manage public debt.

18. The authorities remain committed to improving Chad's debt sustainability and consider that improving the non-oil economy's growth performance is key in this regard. A major roadblock for the authorities' well-placed emphasis on developing the non-oil economy is the growing difficulty of attaining concessional borrowing. The authorities are convinced that the economic returns

to projects like electrification and transportation are high enough to justify non-concessional terms and would in turn help further improve debt sustainability. However, their commitment to the program and conservative debt management remains the priority.

Figure 1. Chad: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2019–2029



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices [2/]	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.3%	1.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	17	20
Avg. grace period	5	6

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

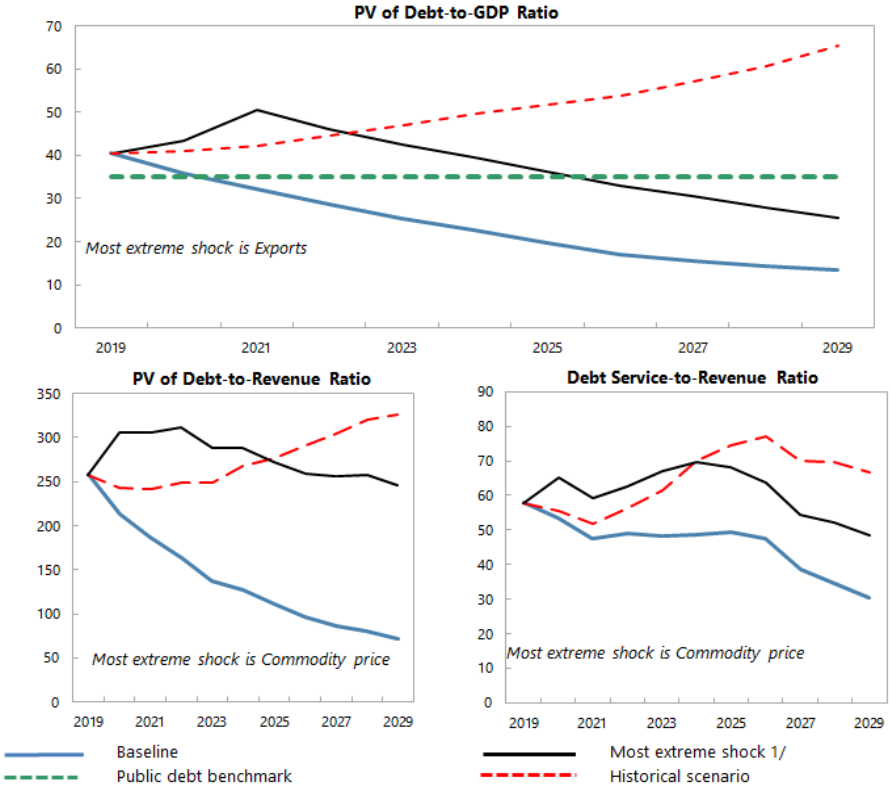
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

The tailored commodity price stress test presented here does not account for the contingency mechanisms in the Glencore debt as Text Figure 1 does.

Figure 2. Chad: Indicators of Public Debt Under Alternative Scenarios, 2019–2029



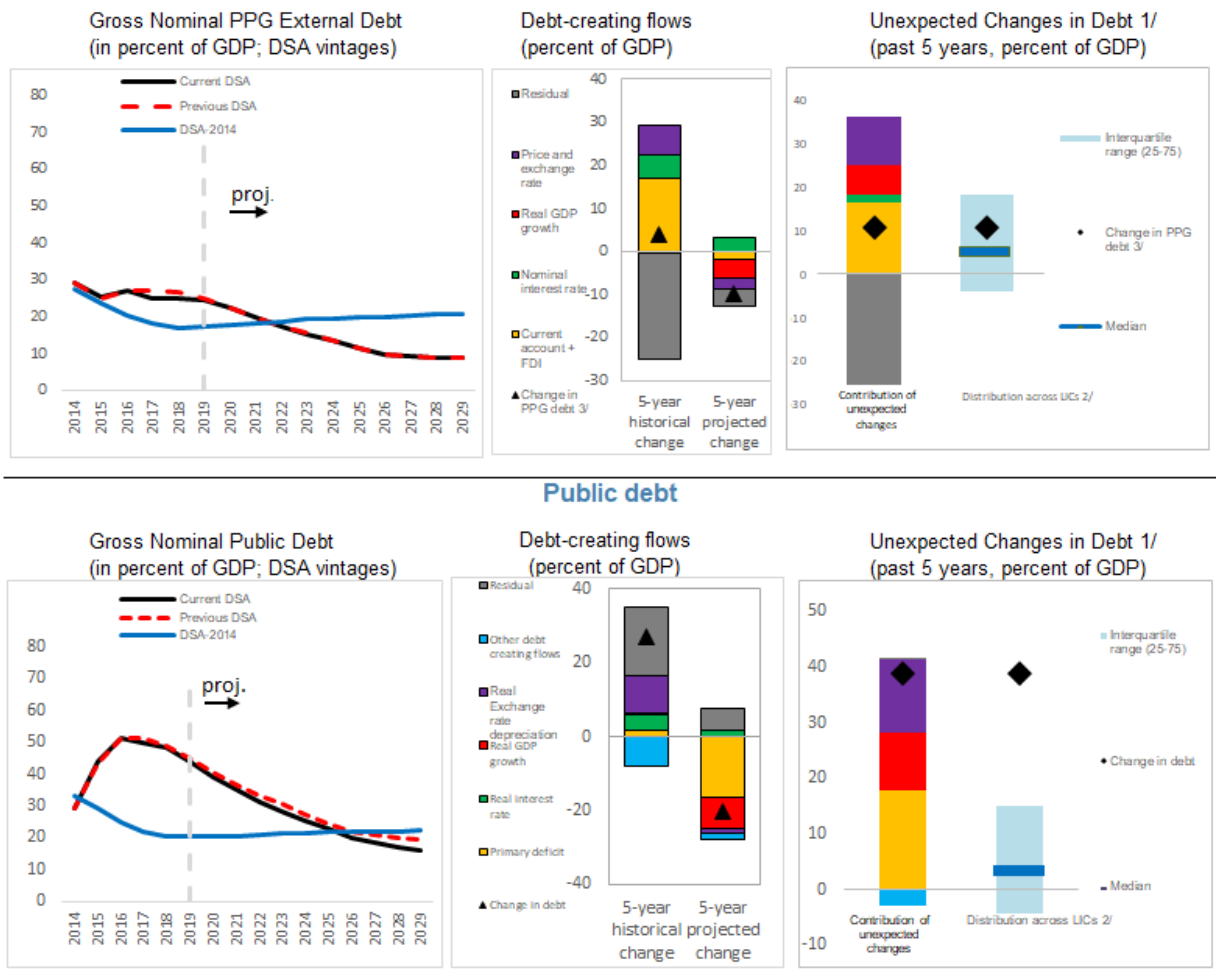
Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	19%	65%
Domestic medium and long-term	21%	20%
Domestic short-term	69%	15%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.3%	1.3%
Avg. maturity (incl. grace period)	17	20
Avg. grace period	5	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	6.0%	4.0%
Avg. maturity (incl. grace period)	3	2
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	2%	3.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 3. Chad: Drivers of Debt Dynamics—Baseline Scenario



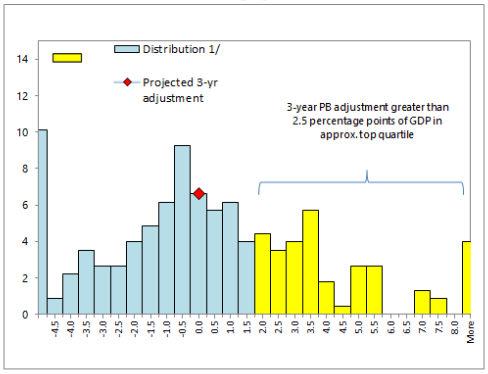
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

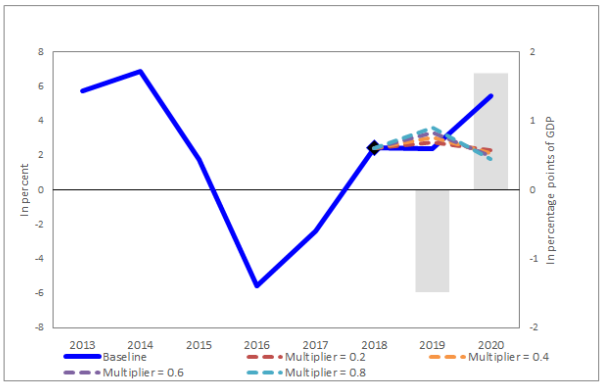
Figure 4. Chad: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



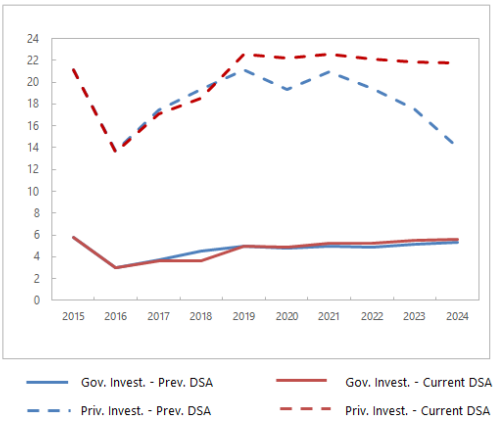
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

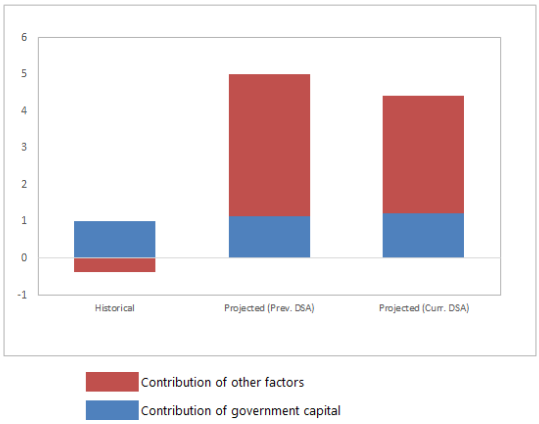
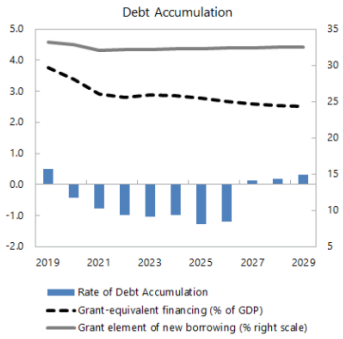


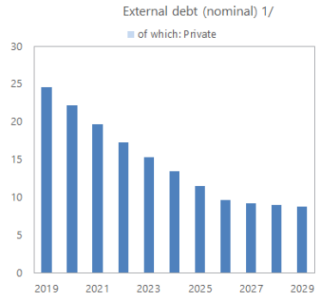
Table 2. Chad: External Debt Sustainability Framework, Baseline Scenario, 2008–2039

(In percent of GDP, unless otherwise indicated)

	Actual											Projections								Average 8/	
	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projection
External debt (nominal) 1/	**16.8**	**26.1**	**24.6**	**21.8**	**20.0**	**21.1**	**29.1**	**25.5**	**27.0**	**25.1**	**25.1**	**24.6**	**22.2**	**19.7**	**17.3**	**15.3**	**13.5**	**8.8**	**9.5**	**24.5**	**14.6**
of which: public and publicly guaranteed (PPG)	**16.8**	**26.1**	**24.6**	**21.8**	**20.0**	**21.1**	**29.1**	**25.5**	**27.0**	**25.1**	**25.1**	**24.6**	**22.2**	**19.7**	**17.3**	**15.3**	**13.5**	**8.8**	**9.5**	**24.5**	**14.6**
Change in external debt	...	9.3	-1.5	-2.8	-1.7	1.1	8.0	-3.7	1.5	-1.9	0.0	-0.5	-2.4	-2.5	-2.5	-1.9	-1.8	-0.2	1.1		
Identified net debt-creating flows	**...**	**3.7**	**0.0**	**-1.7**	**2.7**	**4.2**	**2.2**	**16.6**	**9.6**	**3.3**	**-3.0**	**-0.3**	**-2.2**	**-1.7**	**-2.4**	**-2.0**	**-1.6**	**-0.8**	**2.0**	**3.8**	**-1.0**
Non-interest current account deficit	**-3.7**	**8.0**	**8.4**	**5.4**	**7.1**	**8.5**	**8.2**	**12.5**	**8.5**	**5.5**	**2.7**	**5.7**	**5.2**	**5.6**	**4.6**	**4.6**	**4.9**	**2.1**	**3.0**	**7.5**	**4.4**
Deficit in balance of goods and services	-1.7	11.5	10.8	7.5	9.8	9.6	12.5	16.3	15.9	14.6	9.9	11.7	10.1	9.9	8.4	8.4	8.8	5.4	-19.0	**11.9**	**8.1**
Exports	42.5	35.2	37.8	40.6	38.2	33.4	31.5	26.5	23.2	27.0	33.9	32.8	34.7	33.9	34.2	34.0	33.3	30.9	19.0		
Imports	40.9	46.8	48.6	48.1	48.0	43.1	43.9	42.9	39.1	41.7	43.8	44.5	44.8	43.7	42.6	42.4	42.0	36.4	0.0		
Net current transfers (negative = inflow)	-4.4	-7.5	-5.6	-4.3	-4.4	-5.1	-7.9	-7.1	-7.6	-9.4	-8.9	-8.1	-7.4	-6.8	-6.4	-6.3	-6.3	-4.9	-3.9	**-6.8**	**-6.2**
of which: official	-0.7	-0.6	-0.3	-0.3	-0.7	-1.3	-4.0	-2.5	-2.4	-3.1	-3.2	-2.1	-2.0	-1.6	-1.5	-1.5	-1.7	-1.2	-1.2		
Other current account flows (negative = net inflow)	2.4	3.9	3.2	2.2	1.7	3.9	3.6	3.2	0.1	0.3	1.7	2.2	2.5	2.5	2.6	2.5	2.4	1.5	25.9	**2.4**	**2.5**
Net FDI (negative = inflow)	**-5.8**	**-6.5**	**-5.2**	**-4.5**	**-4.7**	**-4.0**	**-5.2**	**-5.1**	**-2.4**	**-3.6**	**-4.2**	**-6.2**	**-6.9**	**-7.0**	**-6.6**	**-6.4**	**-6.3**	**-2.6**	**-0.8**	**-4.5**	**-5.2**
Endogenous debt dynamics 2/	**...**	**2.3**	**-3.2**	**-2.6**	**0.3**	**-0.3**	**-0.8**	**9.3**	**3.6**	**1.4**	**-1.6**	**0.2**	**-0.5**	**-0.4**	**-0.4**	**-0.2**	**-0.1**	**-0.2**	**-0.1**		
Contribution from nominal interest rate	...	0.2	0.2	0.4	0.7	0.6	0.7	1.1	1.7	1.0	0.6	0.7	0.7	0.6	0.5	0.5	0.4	0.1	0.1		
Contribution from real GDP growth	...	-0.8	-3.1	0.0	-1.9	-1.1	-1.4	-0.7	1.5	0.6	-0.6	-0.6	-1.2	-1.0	-1.0	-0.7	-0.5	-0.3	-0.2		
Contribution from price and exchange rate changes	...	2.8	-0.3	-3.0	1.5	0.2	-0.2	8.8	0.4	-0.3	-1.6	...	...	...	...	...	...	...	...		
Residual 3/	**...**	**5.6**	**-1.4**	**-1.2**	**-4.5**	**-3.2**	**5.8**	**-20.3**	**-8.1**	**-5.2**	**3.0**	**-0.2**	**-0.2**	**-0.7**	**0.0**	**0.0**	**-0.2**	**0.6**	**-0.9**	**-2.9**	**-0.5**
of which: exceptional financing	...	0.0	0.0	0.0	0.0	0.0	-0.1	-0.8	-1.1	-1.0	-1.3	-1.6	-0.7	-0.3	-0.3	-0.3	-0.3	-0.2	-0.1		
Sustainability indicators																					
PV of PPG external debt-to-GDP ratio	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**21.3**	**21.5**	**19.2**	**17.0**	**14.6**	**12.8**	**11.0**	**6.6**	**7.0**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**62.9**	**65.5**	**55.3**	**50.1**	**42.8**	**37.6**	**33.1**	**21.3**	**37.1**		
PPG debt service-to-exports ratio	**3.5**	**2.0**	**1.5**	**2.2**	**3.2**	**3.9**	**15.6**	**9.5**	**14.2**	**8.8**	**6.0**	**4.5**	**6.4**	**6.6**	**6.8**	**6.9**	**6.5**	**2.2**	**3.3**		
PPG debt service-to-revenue ratio	**7.1**	**5.7**	**3.1**	**3.9**	**5.7**	**7.1**	**31.0**	**24.0**	**34.5**	**22.5**	**16.9**	**11.7**	**16.0**	**15.4**	**15.5**	**14.9**	**14.3**	**4.0**	**3.9**		
Gross external financing need (Million of U.S. dollars)	...	...	...	...	453.4	755.2	1114.1	1084.2	953.4	434.9	64.4	117.9	66.1	121.7	49.3	92.4	117.3	28.5	1122.9		
Key macroeconomic assumptions																					
Real GDP growth (in percent)	3.1	4.1	13.6	0.1	8.8	5.8	6.9	1.8	-5.6	-2.4	2.4	2.4	5.5	4.8	5.4	4.1	3.7	3.8	3.1	**3.6**	**4.1**
GDP deflator in US dollar terms (change in percent)	16.6	-14.3	1.2	13.7	-6.4	-1.0	0.8	-23.2	-1.4	1.2	7.0	-0.7	4.0	3.6	3.5	2.1	3.5	2.9	2.9	**-2.2**	**2.8**
Effective interest rate (percent) 4/	...	1.3	0.7	1.8	3.3	3.4	3.7	3.1	6.2	3.8	2.8	3.0	3.1	2.7	2.9	2.8	2.7	1.3	1.3	**3.0**	**2.4**
Growth of exports of G&S (US dollar terms, in percent)	15.3	-26.1	23.3	22.3	-4.1	-8.4	1.4	-34.0	-18.7	15.4	37.6	-1.8	16.0	6.0	10.2	5.7	5.0	5.4	1.5	**0.9**	**6.2**
Growth of imports of G&S (US dollar terms, in percent)	19.3	2.1	19.4	12.7	1.6	-6.0	9.9	-23.7	-15.1	5.3	15.3	3.1	10.6	6.0	6.3	5.7	6.5	5.3	...	**2.2**	**5.2**
Grant element of new public sector borrowing (in percent)		...	...	...	...	...	...	...	...	...	...	33.3	32.8	32.1	32.2	32.2	32.3	32.5	32.5	**...**	**32.4**
Government revenues (excluding grants, in percent of GDP)	21.2	12.3	18.9	23.2	21.7	18.5	15.8	10.5	9.5	10.6	12.0	12.7	13.8	14.5	14.9	15.7	15.2	16.8	16.2	**15.3**	**15.1**
Aid flows (in Million of US dollars) 5/	169.2	350.6	263.4	279.6	477.1	377.5	369.1	433.3	326.8	527.7	480.2	364.8	384.6	363.4	381.3	414.5	442.6	533.6	799.6		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	...	...	...	...	...	...	...	...	3.8	3.4	2.9	2.8	2.9	2.9	2.5	2.4	**...**	**2.9**
Grant-equivalent financing (in percent of external financing)	...	...	...	...	...	...	...	...	...	...	...	71.3	80.3	81.9	81.7	81.2	80.9	79.5	63.1	**...**	**79.8**
Nominal GDP (Million of US dollars)	10,438	9,312	10,702	12,181	12,411	12,994	14,003	10,952	10,202	10,078	11,051	11,238	12,330	13,385	14,600	15,516	16,663	23,203	40,119		
Nominal dollar GDP growth	...	-10.8	14.9	13.8	1.9	4.7	7.8	-21.8	-6.8	-1.2	9.7	1.7	9.7	8.5	9.1	6.3	7.4	6.8	6.1	**1.2**	**7.0**
Memorandum items:																					
PV of external debt 7/	...	...	...	...	...	...	...	...	...	...	21.3	21.5	19.2	17.0	14.6	12.8	11.0	6.6	7.0		
In percent of exports	...	...	...	...	...	...	...	...	...	...	62.9	65.5	55.3	50.1	42.8	37.6	33.1	21.3	37.1		
Total external debt service-to-exports ratio	3.5	2.0	1.5	2.2	3.2	3.9	15.6	9.5	14.2	8.8	6.0	4.5	6.4	6.6	6.8	6.9	6.5	2.2	3.3		
PV of PPG external debt (in Million of US dollars)											2358.6	2413.4	2363.8	2269.7	2137.0	1983.4	1832.1	1532.9	2824.8		
(PVt-PVt-1)/GDPt-1 (in percent)												0.5	-0.4	-0.8	-1.0	-1.1	-1.0	0.3	1.2		
Non-interest current account deficit that stabilizes debt ratio	...	-1.3	9.8	8.3	8.8	7.4	0.2	16.1	7.0	7.4	2.7	6.2	7.6	8.1	7.1	6.6	6.7	2.3	1.9		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

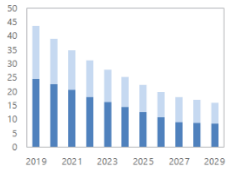
Table 3. Chad: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008–2039

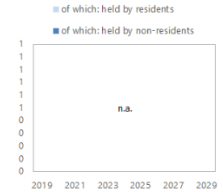
(In percent of GDP, unless otherwise indicated)

	Actual								Projections										Average 6/	
	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**16.8**	**26.1**	**24.6**	**21.8**	**20.0**	**21.1**	**29.1**	**43.8**	**51.4**	**49.7**	**48.2**	**43.8**	**39.0**	**34.9**	**31.1**	**28.0**	**25.4**	**16.1**	**14.3**	**33.6**	**26.9**
of which: external debt	**16.8**	**26.1**	**24.6**	**21.8**	**20.0**	**21.1**	**29.1**	**25.5**	**27.0**	**25.1**	**25.1**	**24.6**	**22.2**	**19.7**	**17.3**	**15.3**	**13.5**	**8.8**	**9.5**	**24.5**	**14.6**
Change in public sector debt	...	9.3	-1.5	-2.8	-1.7	1.1	8.0	14.7	7.6	-1.7	-1.5	-4.5	-4.8	-4.1	-3.8	-3.2	-2.6	-1.0	0.3		
Identified debt-creating flows		**8.8**	**1.4**	**-4.5**	**-2.2**	**0.6**	**0.6**	**9.3**	**5.4**	**-3.1**	**-3.8**	**-3.4**	**-5.9**	**-5.4**	**-5.6**	**-5.7**	**-5.0**	**-5.2**	**-18.4**	**1.2**	**-5.1**
Primary deficit	**-3.9**	**8.7**	**3.6**	**-3.0**	**-0.9**	**1.5**	**3.6**	**2.7**	**-0.1**	**-1.3**	**-3.0**	**-1.5**	**-3.2**	**-3.2**	**-3.9**	**-4.6**	**-3.9**	**-4.5**	**-18.0**	**1.2**	**-3.7**
Revenue and grants	22.4	14.9	20.2	24.8	24.4	20.7	17.8	14.0	12.4	14.6	15.3	15.7	16.8	17.1	17.4	18.3	17.8	19.0	18.0	**17.9**	**17.6**
of which: grants	1.2	2.6	1.3	1.6	2.6	2.3	2.0	3.4	2.9	4.0	3.2	3.0	3.0	2.6	2.5	2.6	2.5	2.2	1.8		
Primary (noninterest) expenditure	18.5	23.7	23.8	21.8	23.4	22.3	21.4	16.7	12.4	13.3	12.2	14.2	13.6	13.9	13.6	13.7	13.8	14.5	0.0	**19.1**	**13.9**
Automatic debt dynamics		**0.1**	**-2.1**	**-1.5**	**-1.3**	**-0.5**	**1.4**	**7.3**	**5.9**	**0.0**	**-0.2**	**-1.4**	**-2.3**	**-1.8**	**-1.5**	**-0.8**	**-0.7**	**-0.6**	**-0.4**		
Contribution from interest rate/growth differential	...	-0.6	-3.2	-0.1	-1.5	-0.8	-1.0	0.1	4.4	2.4	-1.2	-0.9	-1.9	-1.6	-1.3	-0.8	-0.6	-0.5	-0.3		
of which: contribution from average real interest rate	...	0.1	-0.1	-0.1	0.3	0.3	0.3	0.6	1.8	1.2	0.0	0.2	0.3	0.2	0.5	0.4	0.4	0.1	0.1		
of which: contribution from real GDP growth	...	-0.7	-3.1	0.0	-1.8	-1.1	-1.4	-0.5	2.6	1.3	-1.2	-1.1	-2.3	-1.8	-1.8	-1.2	-1.0	-0.6	-0.4		
Contribution from real exchange rate depreciation		0.6	1.2	-1.4	0.2	0.3	2.5	7.2	1.5	-2.4	1.0	...	...	...	...	...	...	...	...		
Other identified debt-creating flows		**0.0**	**-0.2**	**0.0**	**-0.1**	**-0.5**	**-4.4**	**-0.7**	**-0.5**	**-1.8**	**-0.6**	**-0.4**	**-0.4**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.2**	**-0.1**	**-0.9**	**-0.3**
Privatization receipts (negative)		0.0	-0.2	0.0	-0.1	0.0	-4.0	-0.4	0.0	-1.2	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)		0.0	0.0	0.0	0.0	-0.5	-0.4	-0.3	-0.5	-0.5	-0.4	-0.4	-0.4	-0.3	-0.3	-0.3	-0.3	-0.2	-0.1		
Other debt creating or reducing flow (please specify)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual		**0.5**	**-2.8**	**1.6**	**0.5**	**0.5**	**7.4**	**5.4**	**2.2**	**1.4**	**2.4**	**-1.5**	**0.8**	**1.0**	**1.7**	**2.6**	**2.2**	**4.2**	**18.7**	**1.9**	**2.0**
Sustainability indicators																					
PV of public debt-to-GDP ratio 2/	...	...	...	...	...	...	...		...	...	**45.3**	**40.5**	**35.9**	**32.1**	**28.5**	**25.4**	**22.9**	**13.9**	**11.9**		
PV of public debt-to-revenue and grants ratio	...	...	...	...	...	...	...		...	...	**296.7**	**258.2**	**213.6**	**187.4**	**163.5**	**139.0**	**129.0**	**73.1**	**65.9**		
Debt service-to-revenue and grants ratio 3/		**4.7**	**2.9**	**3.6**	**5.0**	**6.4**	**27.6**	**22.6**	**39.2**	**46.5**	**18.3**	**57.8**	**53.4**	**47.5**	**49.1**	**48.7**	**49.2**	**32.3**	**28.7**		
Gross financing need 4/	-2.5	9.2	4.0	-2.5	-0.4	2.2	7.8	4.5	2.9	4.1	-0.2	7.3	6.2	5.4	5.9	5.4	5.8	1.8	4.6		
Key macroeconomic and fiscal assumptions																					
Real GDP growth (in percent)	3.1	4.1	13.6	0.1	8.8	5.8	6.9	1.8	-5.6	-2.4	2.4	2.4	5.5	4.8	5.4	4.1	3.7	3.8	3.1	**3.6**	**4.1**
Average nominal interest rate on external debt (in percent)		1.4	0.6	1.8	3.5	3.3	3.6	3.3	6.3	4.0	2.7	3.1	3.1	2.7	2.9	2.8	2.7	1.3	1.3	**3.1**	**2.4**
Average real interest rate on domestic debt (in percent)	0	12.0	-5.2	-6.2	2.2	7.8	2.9	13.3	2.1	2.3	-0.5	-0.6	0.4	0.9	3.0	2.5	3.2	2.0	2.8	**3.1**	**1.6**
Real exchange rate depreciation (in percent, + indicates depreciation)		3.9	5.2	-5.5	1.0	1.6	12.4	24.5	5.5	-8.5	3.9	...	...	...	...	...	...	...	...	**4.4**	**...**
Inflation rate (GDP deflator, in percent)	8.6	-9.5	6.2	8.4	1.3	-4.2	0.8	-8.0	-1.1	-0.8	2.3	2.6	2.9	3.0	2.9	1.7	2.9	2.9	2.9	**-0.5**	**2.8**
Growth of real primary spending (deflated by GDP deflator, in percent)		32.9	14.3	-8.5	17.2	0.5	2.7	-20.8	-29.9	5.0	-5.7	18.4	1.1	7.5	2.6	4.9	4.8	6.3	3.9	**0.8**	**5.8**
Primary deficit that stabilizes the debt-to-GDP ratio 5/		...	...	...	...	...	...	...	-7.7	0.4	-1.6	2.9	1.6	0.9	-0.1	-1.5	-1.4	-3.5	-18.3	**-2.9**	**-0.8**
PV of contingent liabilities (not included in public sector debt)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, central bank. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 4. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019–2029

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	21.5	19.2	17.0	14.6	12.8	11.0	9.1	7.4	7.0	6.7	6.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	21.5	23.4	25.0	25.7	27.1	28.4	29.5	29.2	**31.3**	**33.9**	**36.5**
B. Bound Tests											
B1. Real GDP growth	21.5	21.6	21.2	18.3	16.0	13.8	11.4	9.2	8.8	8.4	8.3
B2. Primary balance	21.5	22.2	23.7	21.7	19.5	17.0	14.3	12.1	11.2	10.3	9.7
B3. Exports	21.5	28.2	**39.1**	**35.7**	**33.3**	**30.7**	28.2	25.8	24.5	22.4	20.7
B4. Other flows 3/	21.5	22.1	22.4	19.9	17.9	16.0	14.0	12.1	11.5	10.7	10.1
B6. One-time 30 percent nominal depreciation	21.5	24.0	18.0	15.3	13.0	10.8	8.5	6.5	6.1	6.0	6.1
B6. Combination of B1-B5	21.5	**30.6**	**30.6**	27.4	24.9	22.5	20.0	17.7	16.5	15.3	14.3
C. Tailored Tests											
C1. Combined contingent liabilities	21.5	25.1	23.8	21.4	18.9	16.2	13.5	11.3	10.5	9.9	9.5
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	21.5	23.6	25.6	22.9	20.6	18.2	15.5	13.1	11.9	10.6	9.5
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	65.5	55.3	50.1	42.8	37.6	33.1	27.8	22.9	22.1	21.5	21.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	65.5	67.5	73.8	75.1	79.6	85.4	90.2	90.9	98.8	108.3	118.0
B. Bound Tests											
B1. Real GDP growth	65.5	55.3	50.1	42.8	37.6	33.1	27.8	22.9	22.1	21.5	21.3
B2. Primary balance	65.5	64.0	69.9	63.4	57.5	51.0	43.7	37.6	35.3	33.0	31.4
B3. Exports	65.5	115.2	**214.5**	**193.6**	**181.5**	**171.5**	**159.8**	**149.0**	**143.3**	133.1	124.1
B4. Other flows 3/	65.5	63.8	66.3	58.1	52.7	48.1	42.7	37.7	36.2	34.3	32.8
B6. One-time 30 percent nominal depreciation	65.5	55.3	42.5	35.7	30.5	26.0	20.8	16.0	15.3	15.3	15.7
B6. Combination of B1-B5	65.5	101.7	75.1	103.0	94.4	87.1	78.7	70.9	66.8	62.8	59.6
C. Tailored Tests											
C1. Combined contingent liabilities	65.5	72.5	70.3	62.6	55.6	48.8	41.3	35.1	33.2	31.6	30.7
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	65.5	80.9	87.3	74.9	66.0	57.9	48.9	41.8	38.6	34.8	31.6
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	4.5	6.4	6.6	6.8	6.9	6.5	7.2	6.8	2.8	2.6	2.2
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	4.5	6.9	7.8	8.8	9.4	9.7	**11.3**	**11.5**	6.4	7.2	7.6
B. Bound Tests											
B1. Real GDP growth	4.5	6.4	6.6	6.8	6.9	6.5	7.2	6.8	2.8	2.6	2.2
B2. Primary balance	4.5	6.4	6.7	7.1	7.2	6.9	7.5	7.0	3.6	4.2	3.9
B3. Exports	4.5	9.6	**14.3**	**15.9**	**16.0**	**15.2**	**16.5**	**15.6**	**10.5**	**14.9**	**13.6**
B4. Other flows 3/	4.5	6.4	6.7	7.1	7.2	6.8	7.5	7.0	3.6	4.0	3.5
B6. One-time 30 percent nominal depreciation	4.5	6.4	6.6	6.6	6.8	6.4	7.1	6.7	2.7	1.9	1.6
B6. Combination of B1-B5	4.5	8.3	**10.8**	**11.2**	**11.3**	**10.8**	**11.8**	**11.1**	7.4	7.3	6.5
C. Tailored Tests											
C1. Combined contingent liabilities	4.5	6.4	6.9	7.1	7.2	6.8	7.5	7.0	2.9	2.7	2.3
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	4.5	7.8	8.2	8.5	8.5	7.9	8.5	8.0	4.3	4.9	4.3
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**
Debt service-to-revenue ratio											
Baseline	11.7	**16.0**	**15.4**	**15.5**	**14.9**	**14.3**	**15.4**	**14.2**	5.6	5.1	4.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	11.7	**17.3**	**18.2**	**20.1**	**20.4**	**21.2**	**24.2**	**24.0**	12.9	**14.4**	13.9
B. Bound Tests											
B1. Real GDP growth	11.7	**18.0**	**19.2**	**19.4**	**18.7**	**17.9**	**19.3**	**17.8**	7.1	6.4	5.0
B2. Primary balance	11.7	**16.0**	**15.7**	**16.3**	**15.7**	**15.0**	**16.0**	**14.7**	7.3	8.3	7.1
B3. Exports	11.7	**17.0**	**18.0**	**19.6**	**18.7**	**17.9**	**19.0**	**17.6**	11.5	**15.9**	13.5
B4. Other flows 3/	11.7	**16.0**	**15.7**	**16.2**	**15.5**	**14.9**	**15.9**	**14.7**	7.4	7.9	6.4
B6. One-time 30 percent nominal depreciation	11.7	**20.1**	**19.3**	**19.1**	**18.3**	**17.6**	**18.9**	**17.5**	6.8	4.8	3.6
B6. Combination of B1-B5	11.7	**18.1**	**19.7**	**19.9**	**19.0**	**18.2**	**19.5**	**18.0**	11.7	11.2	9.3
C. Tailored Tests											
C1. Combined contingent liabilities	11.7	**16.0**	**16.1**	**16.3**	**15.6**	**15.0**	**15.9**	**14.6**	6.0	5.4	4.2
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	11.7	**20.7**	**20.6**	**21.7**	**19.6**	**18.1**	**18.4**	**16.2**	8.5	9.5	7.6
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 5. Chad: Sensitivity Analysis for Key Indicators of Public Debt, 2019–2029

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	**40.5**	**35.9**	32.0	28.1	25.1	22.3	19.5	16.8	15.3	14.2	13.2
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	**41**	**41**	**42**	**44**	**47**	**49**	**51**	**53**	**57**	**60**	**65**
B. Bound Tests											
B1. Real GDP growth	**41**	**45**	**48**	**46**	**45**	**45**	**43**	**42**	**42**	**43**	**44**
B2. Primary balance	**41**	**42**	**44**	**40**	**36**	34	31	28	26	24	23
B3. Exports	**41**	**43**	**50**	**46**	**42**	**39**	**36**	33	30	28	25
B4. Other flows 3/	**41**	**39**	**37**	33	30	27	24	22	20	18	17
B6. One-time 30 percent nominal depreciation	**41**	**45**	**40**	**35**	32	28	25	21	18	16	15
B6. Combination of B1-B5	**41**	**42**	**41**	34	31	28	27	24	23	22	21
C. Tailored Tests											
C1. Combined contingent liabilities	**41**	**48**	**43**	**38**	**35**	33	30	27	25	23	22
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	**41**	**43**	**44**	**45**	**47**	**48**	**47**	**46**	**46**	**46**	**47**
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Public debt benchmark	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**
PV of Debt-to-Revenue Ratio											
Baseline	258.2	213.6	187.0	161.2	136.2	124.8	108.7	95.1	85.2	78.7	69.7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	258	242	241	246	247	266	275	289	302	319	325
B. Bound Tests											
B1. Real GDP growth	258	263	271	255	237	240	234	231	229	232	224
B2. Primary balance	258	252	259	227	198	187	170	156	143	134	119
B3. Exports	258	258	294	262	230	218	200	185	169	154	133
B4. Other flows 3/	258	231	219	191	164	153	136	122	110	101	88
B6. One-time 30 percent nominal depreciation	258	270	235	205	175	161	140	120	104	92	77
B6. Combination of B1-B5	258	252	236	191	165	158	147	137	128	121	109
C. Tailored Tests											
C1. Combined contingent liabilities	258	284	251	220	192	182	165	151	139	130	116
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	258	306	306	309	286	286	271	257	254	255	245
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	57.8	53.4	47.1	46.9	45.9	45.7	46.6	45.7	36.6	32.8	29.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	58	56	51	54	59	66	71	74	67	67	64
B. Bound Tests											
B1. Real GDP growth	58	59	58	61	61	61	63	63	54	51	47
B2. Primary balance	58	53	58	65	54	47	45	44	36	35	31
B3. Exports	58	53	48	49	48	47	48	47	41	41	36
B4. Other flows 3/	58	53	47	47	46	46	47	46	38	35	31
B6. One-time 30 percent nominal depreciation	58	52	49	47	47	47	48	47	35	31	26
B6. Combination of B1-B5	58	53	49	57	51	48	48	47	38	35	31
C. Tailored Tests											
C1. Combined contingent liabilities	58	53	69	59	49	45	44	43	35	32	28
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	58	65	59	60	64	66	65	61	52	50	47
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.